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To Whom It May Concern,
I am delighted to have the opportunity to enthusiastically recommend Alex Marshall for the Hamilton Award. I have come to know Alex over the past four years since he joined our group as a graduate student in 2022, even before enrolling at UNC. Alex started in the group with a bang – immediately taking on one of our most challenging experiments – developing a new volumetric light sheet microscopy system for our biophysical and biochemical studies of live cells. We have emerged from an unusual period in the lab, with our two senior graduate students working off site (and out of state) this past year while we welcomed into the lab three new graduate students, including Alex. This has meant that Alex and his cohort have not had the mentorship of senior students during their initial ramp up period in the lab. This cold launching has been embraced enthusiastically by Alex, and he has stepped up mightily to take on his challenging project.

Within the research conducted in my own group, Alex has chosen to step into our most challenging project, advanced instrumentation for studying the biophysics of live biological cells. Living cells and tissues are complicated materials that challenge our understanding of even the most basic processes in man-made materials. They are like sponges – they have a matrix of molecules, proteins and polymers, all submerged in a fluid that pervades the sponge. This fluid carries smaller proteins and particles that can move through th sponge – depending on their size. If they are smaller than the holes in the sponge – the particles move freely and flow with he movement of the fluid. If they are larger than the sponge holes, the particles get stuck. In between sizes move more or less freely with the larger particles being slowed down by the sponge. In materials science, these materials are called poroelastic. They are elastic because the material can deform as it is squeezed. They are called "poro – " because of the pores in the material that allow fluid to flow. The interesting thing about these materials is that, of necessity, if an external force is applied - the material will deform and the fluid will flow. In living tissues and cells, the "sponge" is made up of biopolymers such as actin filaments, microtubules, extracellular matrix, and within the cell nucleus, DNA. The fluid that pervades the biomatrix caries small molecules, proteins of various sizes and vesicles. These move within the biomatrix according to their size, with the largest of them having their movement severely restricted. Within the nucleus, it is the movement and distribution of proteins and their complexes called transcription factors that dictate the generation of specific proteins through transcription. Our hypotheses is that dynamic strains on tissues and cells results in the deformation of the nucleus which causes fluid flow that redistributes transcription factors (TF) and therefore alters the proteins that are generated in the cell. This would be a novel insight into how cells respond to external mechanical stresses such as occur in muscles, in joints, and traumatic injury.

Remarkably, there very few studies of these phenomena in man-made materials, and even fewer in live cells. This is the heart of Alex's studies. He has spent the first three years of his lab work developing a novel light sheet microscope for his studies, building the data analysis tools to probe the data he acquires and developing the hydrogel specimens to study stress-induced fluid flow and solute redistribution. These are the first studies at the single particle level of this phenomenon, unique to poroelastic materials. Alex has uncovered the first signs of asymmetric transport during dynamic compressions of hydrogels which will provide the first molecular scale understanding of strain pumping of proteins. This will have a dramatic impact on tissue engineering, on the treatment of joint disorders and provide a new mechanism for the cell-level sensing of mechanical stresses.

Coming from a physics mindset, Alex seeks to understand this process through a predictive theoretical physics model that combines an understanding of physical chemistry and biology. To create this model, he first needs to acquire rapid volume scans of hydrogel materials during dynamic strains, and executing time-synchronized single particle tracking of thousands of mobile nanoparticles. This he has accomplished with little help from senior staff. He is now putting together his work into two publications on novel poroelastic phenomena in hydrogel materials. These will be high impact manuscripts. Alex has planned his studies to be immediately transferrable to experiments on live cells, and is actively working with our collaborator Prof. Kerry Bloom to begin those experiments. He has also started analyzing data from the Bloom lab of the mobility of individually labeled histone proteins to understand their dynamics within the nucleus. His next steps will be to apply dynamic strain to living cell nuclei while tracking the histones, labeled transcription factors and other proteins.

Alex represents a new generation of scientist that knows no disciplinary boundaries, but brings instincts for framing understanding from his starting mindset: physics. Merging fields from physics to materials science to physical chemistry to biology and instrumentation, Alex is endeavoring to do it all! He has accomplished the first leg of this journey: to develop advanced volumetric microscopy methods and instrumentation, and to execute these on hydrogel materials. Alex has accomplished all of this single handedly in my group. While in past years my starting students have come into the lab with the mentorship of senior students, Alex has not had this luxury. I am thrilled that he is bringing his eclectic mind to my group.

Alex is bright, creative and an incisive thinker. Most important, he is highly coachable and is taking personal responsibility for his training in optics, microscopy and cell biology. There is no more important trait for a researcher than their devotion to learning and to teaching themselves. The most successful students are voracious in reading the scientific literature and identifying their own knowledge gaps – and filling these in rapidly. Alex is wonderful at doing these. On the personal side, he is a collegial lab mate with a ready smile and laugh, and generous with his time for mentoring and collaborating. He has a long record in mentoring and teaching undergraduate students that is most impressive. He has assembled a mini-group around him, coaching 9 undergraduates in total, with 4 of them currently active in the group. I am especially impressed with his appreciation of the ways in which he has been mentored across generations of family and friends, and how that has instilled in him a deep enthusiasm - and a sense of obligation – for his outreach activities. He is an enthusiastic presenter at our many presentations to high school and undergraduate students, and to the booth we set up at the annual NC Science Festival.

Alex is an outstanding example of what the Hamilton Award should support: a clear, physics-based thinker tackling challenges at the forefront of biology while also deeply concerned with the world around him. I give him my highest recommendation.

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